

ALDEN SIO

Singaporean Citizen | Eligible for H-1B1 Visa

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EDUCATION

National University of Singapore

Bachelor of Science in Business Analytics (Computing)

Singapore

Aug. 2022 – Apr. 2026

- GPA: 4.53/5.00
- Teaching Assistant: CS2030, BT1101

WORK EXPERIENCE

Adzymic

AI Lead

Singapore

Sep. 2025 - Present

- Spearheaded the E2E architecture of a multi-agent **AI orchestration platform** automating AdTech workflows
- Designed the underlying data and schema infrastructure and human-in-the-loop validation system with multi-stage checkpoints and enrichment modules, increasing **throughput and validation speed by 55%**
- Automated performance analytics with offline **Insight Agents** that created feedback loops, reducing manual reporting overhead by **70%** and **tripling adaptive optimisation speed**

TikTok

Machine Learning Engineer

Singapore

Jan. 2025 – Jul. 2025

- Architected **Agentic Deep Research** for automated reports, reducing manual editing by **20 hours weekly**
- Developed a parallelised **Topic Modeling Pipeline** for **40M+ videos**, reducing runtime by **70%**
- Established scalable frameworks for data ingestion and synthetic generation, enabling NL/DSL-based asset search to improve model training diversity
- Developed an Attribution Model leveraging engagement and CTR insights to guide Shop-Ads clients on optimising content strategies

Synapxe

AI Engineer

Singapore

May. 2024 – Oct. 2024

- Pioneered a **Self-RAG** and **RAG-Fusion** system using **AWS Bedrock**, boosting response relevancy by **60%** and slashing hallucination rates from **40%** to **2%**
- Engineered a multi-modal data ingestion pipeline with **Kedro** to save over **150 annual hours** in manual effort
- Integrated **GraphRAG** for extensive context understanding, constructing a **50k** entity knowledge graph from unstructured data

Panda Commercial Bank PLC.

Software Engineer

Phnom Penh, Cambodia

May. 2023 – Aug. 2023

- Implemented and deployed scalable microservices overseeing **e-Wallet transactions & e-KYC**
- Devised Data Pipelines for Data Cleaning, Extraction, & Analysis, reducing development time by **25%**

PROJECTS

Agent Visualiser (GitHub) | LangChain, LangGraph, Milvus, Neo4j, Chainlit

Sep. 2024 – Oct. 2024

- Built an E2E Agentic-Hybrid-RAG Chatbot to help my colleagues visualise Agentic Interactions
- Accelerated team productivity by **30%** by streamlining knowledge transfer and reducing learning curves

FashoReco (GitHub) | pandas, sklearn, word2vec, scaling, clustering

Jan. 2024 – May. 2024

- Developed an ensemble Recommendation System using a Time-Infused Collaborative Model, and a Time-Decay Content Based Model, improving recommendation hit rate from **0.03%** to **0.20%**
- Employed Multi-Dimensional Clustering to facilitate in-depth Network Analysis for tailored recommendations

TECHNICAL SKILLS

Languages & Databases: Python, Java, Ruby, C#, R, SQL, PostgreSQL, MySQL, MongoDB, Neo4j

Systems & Developer Tools: LLMs, VLMs, Git, AWS, GCP, Azure, Docker, Postman, Firebase, Tableau

Frameworks & Libraries: PyTorch, Scikit-learn, FastAPI, Flask, LangChain, LangGraph, LlamaIndex, Pydantic-AI